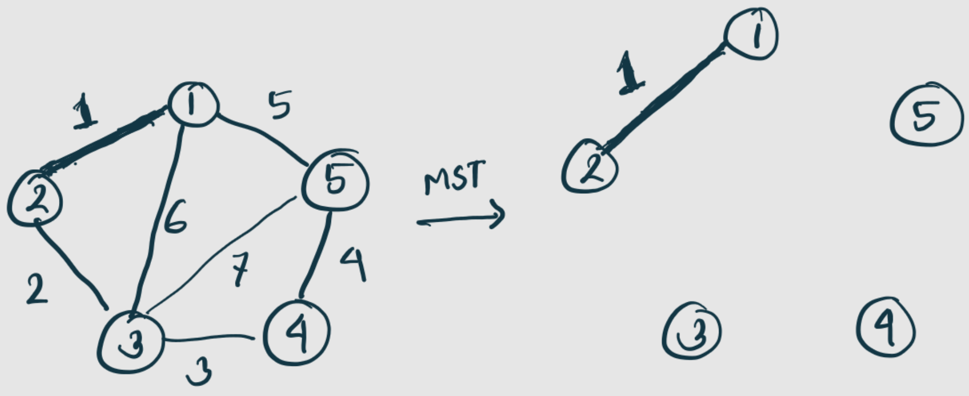
Greedy Algorithm

* smallest / Largest → optimal solution
* Best pick gives optimal solution ✓
* Local optimal = Global optimal ✓

Example: 1. Huffman
2. Fractional Knapsack
3. MST (Kruskull)

* Kruskull



ω	(u,v)
1	(1,2)
⋮	⋮

* Fractional knapsack

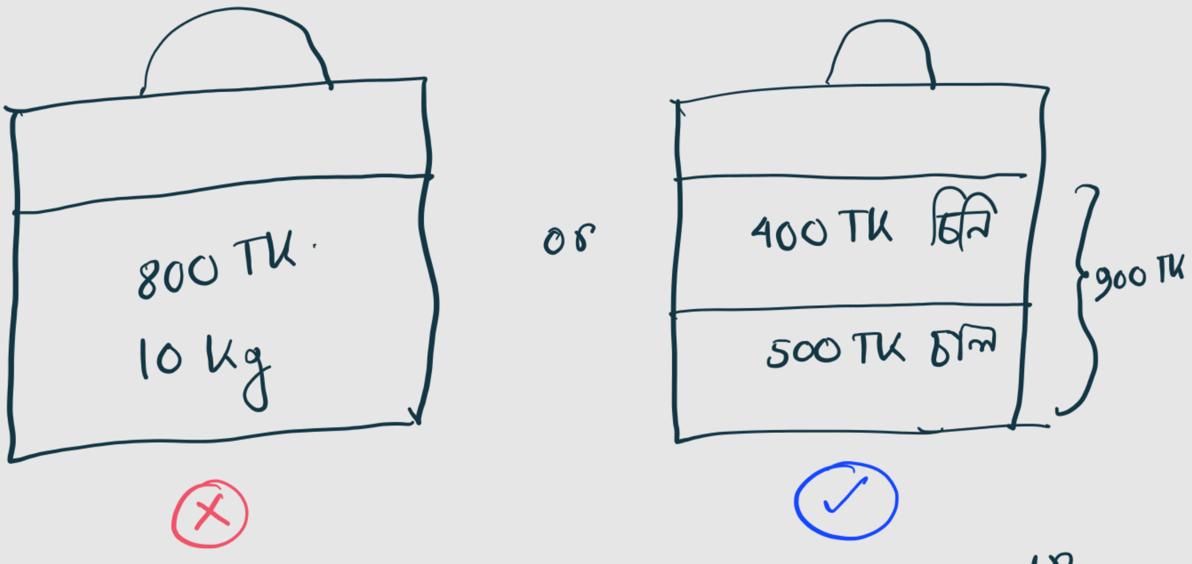
(kg)

চাল	50 kg	2500 TK	50 tk/kg
ডাল	20 kg	4000 TK	200 tk/kg ✓

* 0/1 knapsack

500 TK 5 kg	800 TK ✓ 10 kg	400 TK 3 kg	12 kg
চাল	ডাল	চিনি	

Ans:



Greedy algorithm won't give optimal solⁿ.

Huffman Coding

→ Encoding algorithm

He llo $\xrightarrow{\text{Encode}}$ 101 . . . . . . $\xrightarrow{\text{Decode}}$ He llo

ASCII

* He llo → $6 \times 8 = 48$ bit

* Frequency high → Codeword ছোট করে দেয়

Eyes are happy and hello

e → 3   11100101 → 10

l → 2   10112110 → 1011

* Is huffman coding greedy and does it give optimal solution?

* Same frequency at the end or random position of priority Queue? which is more optimal?

Eerie eyes seen near lake.

(a) Construct the Huffman Tree ✓

(b) Show codeword for each character

(c) Show encoded message

(d) Total number of bits ✓

(e) Decode 101000110111101110 ✓

✓ (f) Show that higher frequencies always get shorter keywords.

✓ (g) If you want to send 'BRACU' with 8 bit encoding scheme how many extra bit would you need?

*Full stop, comma miss করা যাবে না

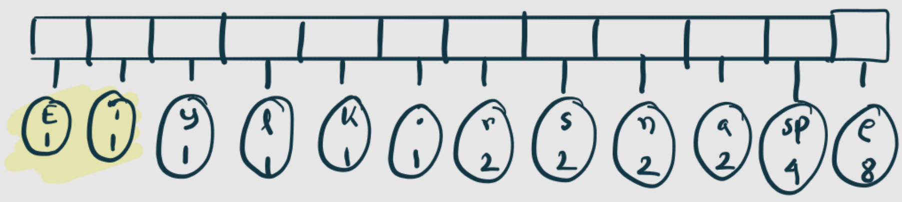
Ans:

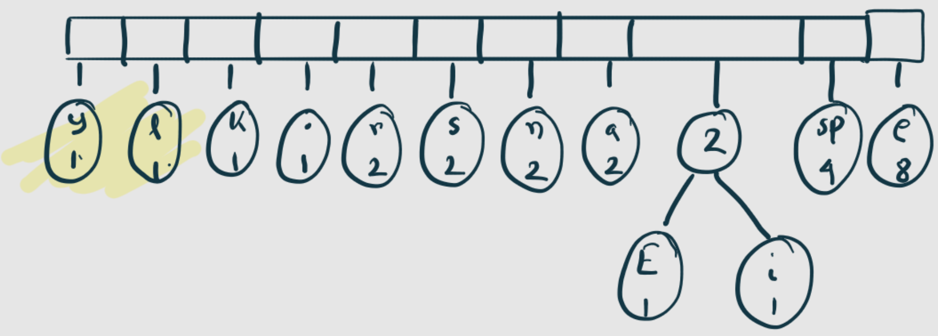
(a)

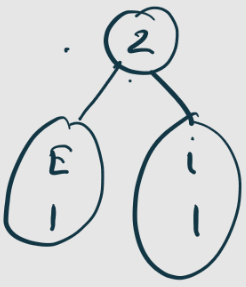
~~Eerie eyes seen near lake~~.

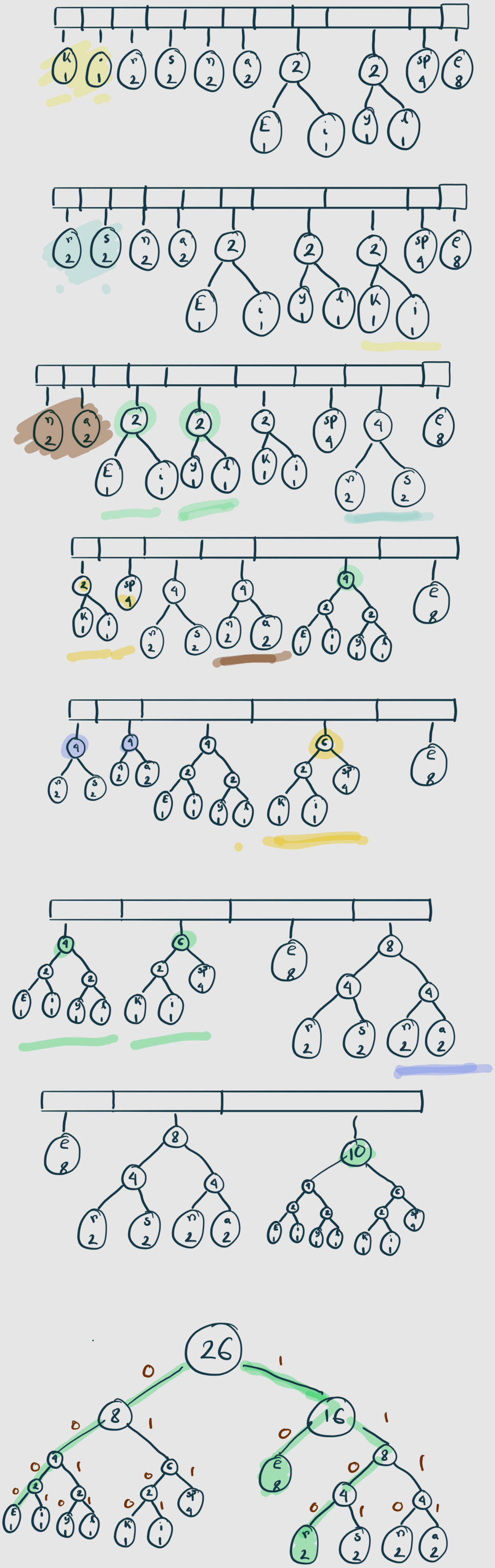
E e r i space
y s n a r l K .

Char	Freq
E	1 ✓
e	8
r	2
space	4
i	1 ✓
y	1 ✓
s	2
n	2
a	2
l	1 ✓
K	1 ✓
.	1 ✓









(b)

Char	Code
E	0000 ✓
i	0001
y	0010
l	0011
k	0100
.	0101
space	011
e	10
r	1100
s	1101
n	1110
a	1111

(c)

Eerie eyes seen near lake.

Encode : 0000 10 1100 0001 10 011

10 0010 10 1101 011 1101 10

1111 011 0011 1111 0100 10 0101

(d)

Without Huffman = 26 × 8 = 208

With huffman = $(4\times1) + (4\times1) + (2\times8) + \ldots$

= $\boxed{}$

Waste = 208 − $\boxed{}$

(e)

101000110111110111 10

ee l sn

Char	Code
E	0000 ✓
i	0001
y	0010
l	0011
k	0100
.	0101
space	011
e	10
r	1100
s	1101
n	1110
a	1111

(g)

BRACU → 8 bit

= 5 × 8 = 40

= $\boxed{}$

Extra = 40 − $\boxed{}$ = $\boxed{}$

Huffman + Knapsack

4 You are designing a data compression and transmission system for a sensor network that monitors environmental conditions. Each type of sensor reading has a frequency of occurrence and an impact factor indicating its importance for critical decision-making.

Your goals are to compress the data efficiently for transmission using a variable-length encoding system and maximize the total impact factor transmitted within the system's data capacity.

Sensor Data Table:

Sensor Reading	Frequency of occurrence	Impact Factor
Temperature	45	10
Humidity	13	6
Pressure	12	8
Light	16	7
CO2	9	5
Wind Speed	5	4
Rainfall	7	6
UV Index	3	3

$\frac{10}{45}$

a. [CO1] **Construct** a Huffman tree for these sensor readings based on their frequencies of occurrence. Then **assign** binary codes to each reading. **05**

b. [CO1] **Calculate** the total number of bits required for the transmission of all readings. **02**

c. [CO1] Suppose, the system can transmit only 15 units of data. Each sensor reading generates 1 (one) unit of data per occurrence. You can partially transmit the readings but its impact factor drops proportionally if you do so. **03**
Determine which readings to transmit and with what frequency, to maximize the total impact factor within the 15-unit data limit. **Calculate** the maximum total impact factor.

Fractional

3 Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

CO1 a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses. **04**

CO1 b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. Note that, selecting a course still means taking the whole with full credits. **02**

0/1

CO1 c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message. **04**

Dynamic Programming (DP)

→ Brute force

* Fibonnacci Series

1, 1, 2, 3, 5, 8, 13, 21, 34 . . .

n-2 n-1 n

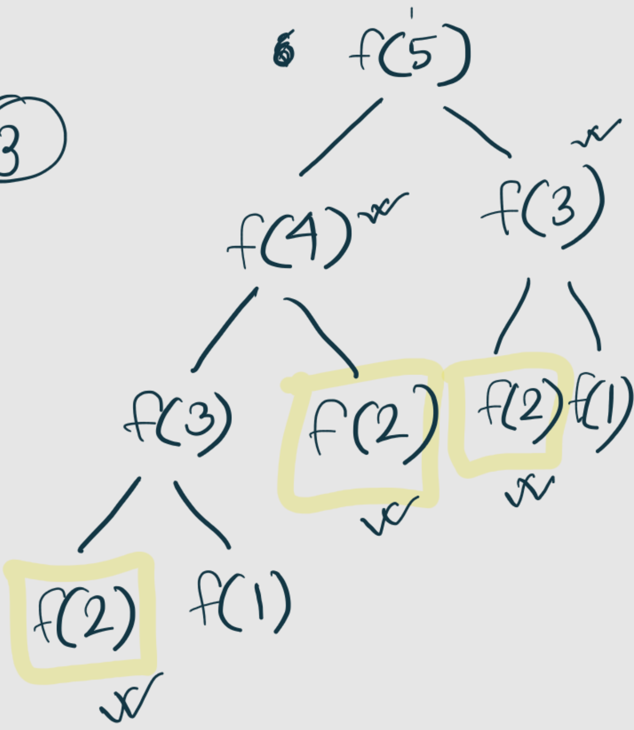
1, 1, 2, 3, 5

1	1	2	3	5

1, 1, 2, 3, 5, 8, 13, 21, 34

Memoization

3




```
int fib(int indx){
  if(indx==1 || indx==2) return 1;
  if(ans[indx] != 0){
      return ans[indx];
  }
  int a = fib(indx-1);
  int b = fib(indx-2);
  sum = a+b;
  ans[indx] = sum;
  retun sum;
}
```

* Top Down/Recursive DP
* Overlapping subproblem (DP)

* Bottom Up / Iterative

1	1	2	3	

```
ans[1] = 1;
ans [2] = 1;

for(i=3; i<=n; i++){
    ans[ i ] = ans[i-1] + ans[i-2]
}
```